Answer Key - Physics - Undergraduate Level Assessment 22 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) decreased by a factor of 9; B) decreased by a factor of 49; C) decreased by a factor of 81; D) increased by a factor of 9
   Note: Correct option: A - decreased by a factor of 9

2. **Answer:** C
   Options: A) 1.2 * 10^9 m/s; B) 3.0 * 10^8 m/s; C) 1.5 * 10^8 m/s; D) 1.0 * 10^8 m/s
   Note: Correct option: C - 1.5 * 10^8 m/s

3. **Answer:** D
   Options: A) mc/2; B) mc/($2^{(}1/2)$); C) mc; D) ($3^{(}1/2)$)mc
   Note: Correct option: D - ($3^{(}1/2)$)mc

4. **Answer:** D
   Options: A) 44 Hz; B) 196 Hz; C) 262 Hz; D) 393 Hz
   Note: Correct option: D - 393 Hz

5. **Answer:** A
   Options: A) a dye laser; B) a helium-neon laser; C) an excimer laser; D) a ruby laser
   Note: Correct option: A - a dye laser

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'gamma rays'.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'rate of change of the electric flux through S'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'rate of change of the electric flux through S'.

10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 1.00032. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** B. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key